

---

---

# University Basketball Analysis

— Mey Siv —

---

---

# University Basketball Analysis

Raw statistics like points or plus-minus can't tell how well players perform together to win the game.

Basketball is more complex than just looking at individual stats - affect by teammates, opponents and playing time.

Question: Do certain pairs of players create more value than their individual ratings would predict?

# Scraping Data from the UniSport

Women's event-level play-by-play data from four different universities in Australia from 2024 to 2025.

- La Trobe (LTU)
- Victoria University (VU)
- Melbourne University (MELB)
- The University of Sydney (USYD)

The data includes player, team identifiers, scoring events, substitutions, game-clock information and match results.

# Approach

## **Player connection:**

- Reconstruct lineups → estimate adjusted pair performance → compute network centrality to identify structurally important players

## **Team strategy:**

- examine how central players are deployed in-game

## **Predictive modelling**

- test whether centrality improves outcome prediction beyond raw playing time.

## Expected results

A ranked table of standout player pairings and an interactive network highlighting key combinations and centrality.

Insight into how coaches deploy high-value players in-game; and evidence-based input for substitution, rotation, and strategy decisions.